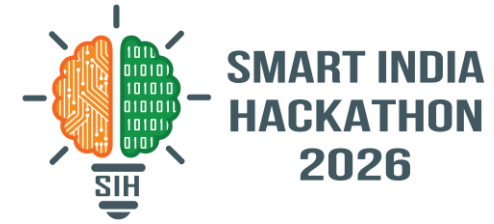
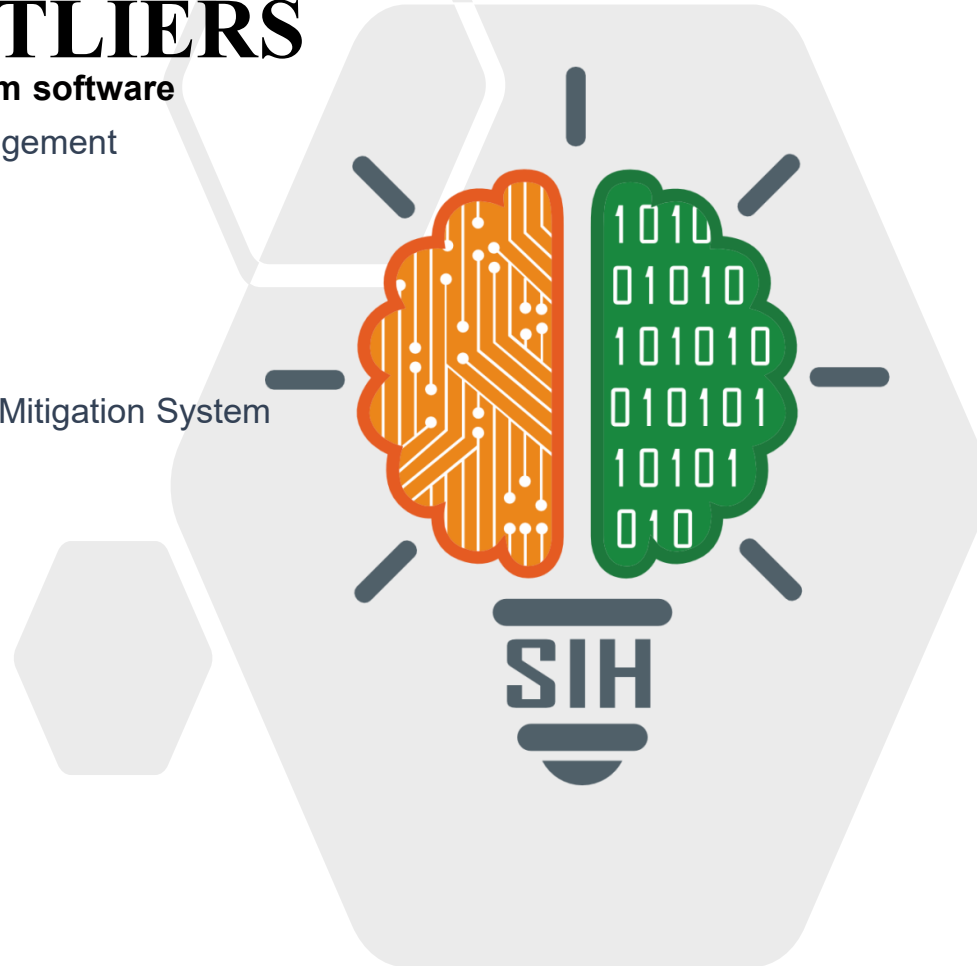


SMART INDIA HACKATHON 2026



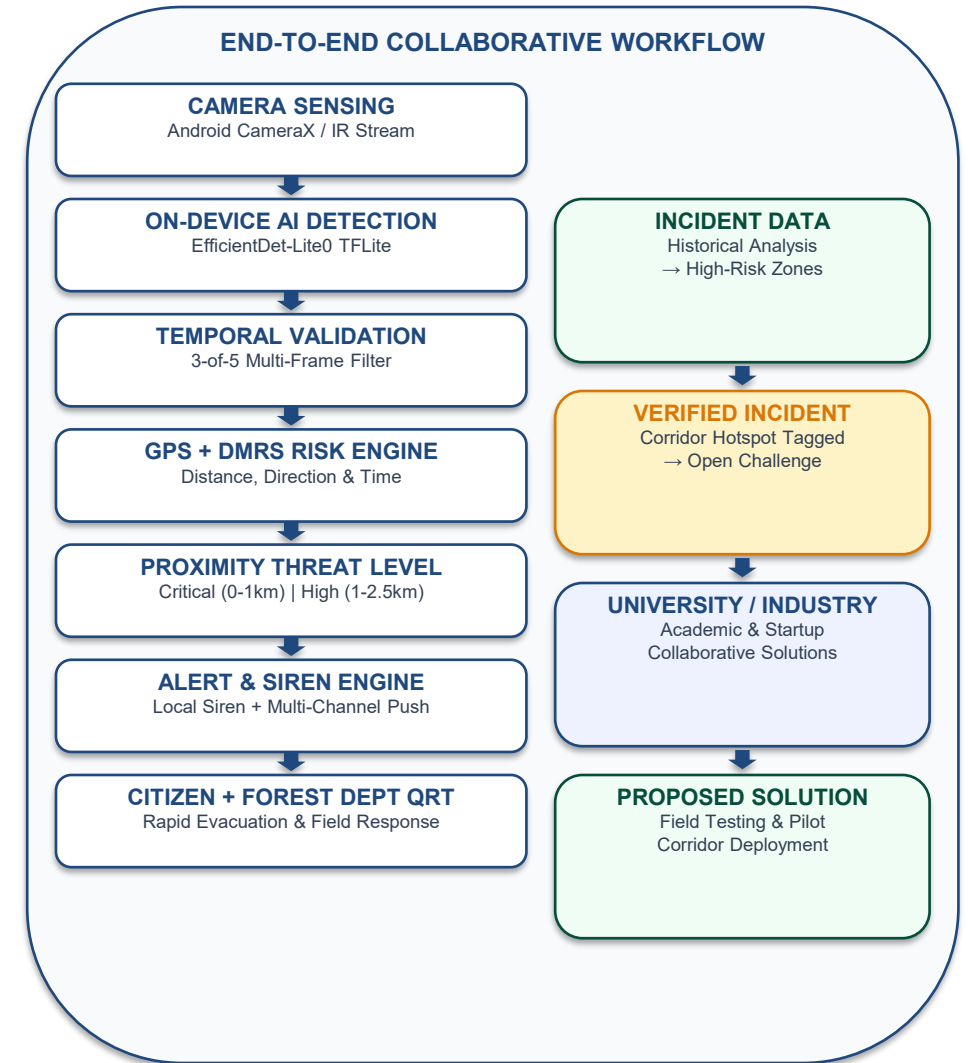
THE OUTLIERS

- **Problem Statement ID** – SIH26043
- **Problem Statement Title** – AI based Elephant attack alert system software
- **Theme** – Smart Automation / Wildlife Conservation & Disaster Management
- **PS Category** – Software
- **Team ID** – [KEEP EXISTING VALUE / PLACEHOLDER]
- **Team Name (Registered on portal)** – The Outliers
- **Project Title** – ELEPHANT GUARD
- **Subtitle** – Smart Elephant Early Warning & Human-Wildlife Conflict Mitigation System



❖ Proposed Solution

- 1. AI-BASED EARLY WARNING:** On-device computer vision detects elephants from live camera feeds and calculates an estimated detection location using device GPS.
- 2. TEMPORAL VALIDATION:** Multiple recent detections are validated (3-of-5 queue) before alert escalation, eliminating single-frame false positives.
- 3. DYNAMIC RISK ASSESSMENT (DMRS):** Risk is calculated using detection confidence, proximity, movement direction, time-to-collision, time of day and historical hotspot records.
- 4. PROXIMITY-BASED ALERTS:**
 - 0–1 km: Critical → emergency mobile push + audible siren
 - 1–2.5 km: High/Warning → strong warning notification
 - >2.5 km: Safe buffer monitoring (no emergency siren)
- 5. FOREST DEPARTMENT RESPONSE:** Real-time tactical map, live incident feed, and QRT dispatch coordination support rapid field officer intervention.
- 6. COLLABORATIVE PROBLEM SOLVING:** Verified corridor incidents can be converted into open challenges for universities and industry partners to propose innovative field solutions.



Technology-

Frontend / Mobile:

- Android Native (Kotlin 2.0 + Jetpack Compose)
- CameraX API for real-time video stream capture
- OpenStreetMap (osmdroid) for local offline GIS mapping

AI / ML Pipeline:

- TensorFlow Lite on-device inference engine
- Quantized EfficientDet-Lite0 object detection model
- Multi-frame temporal validation queue & confidence filter

Backend Architecture:

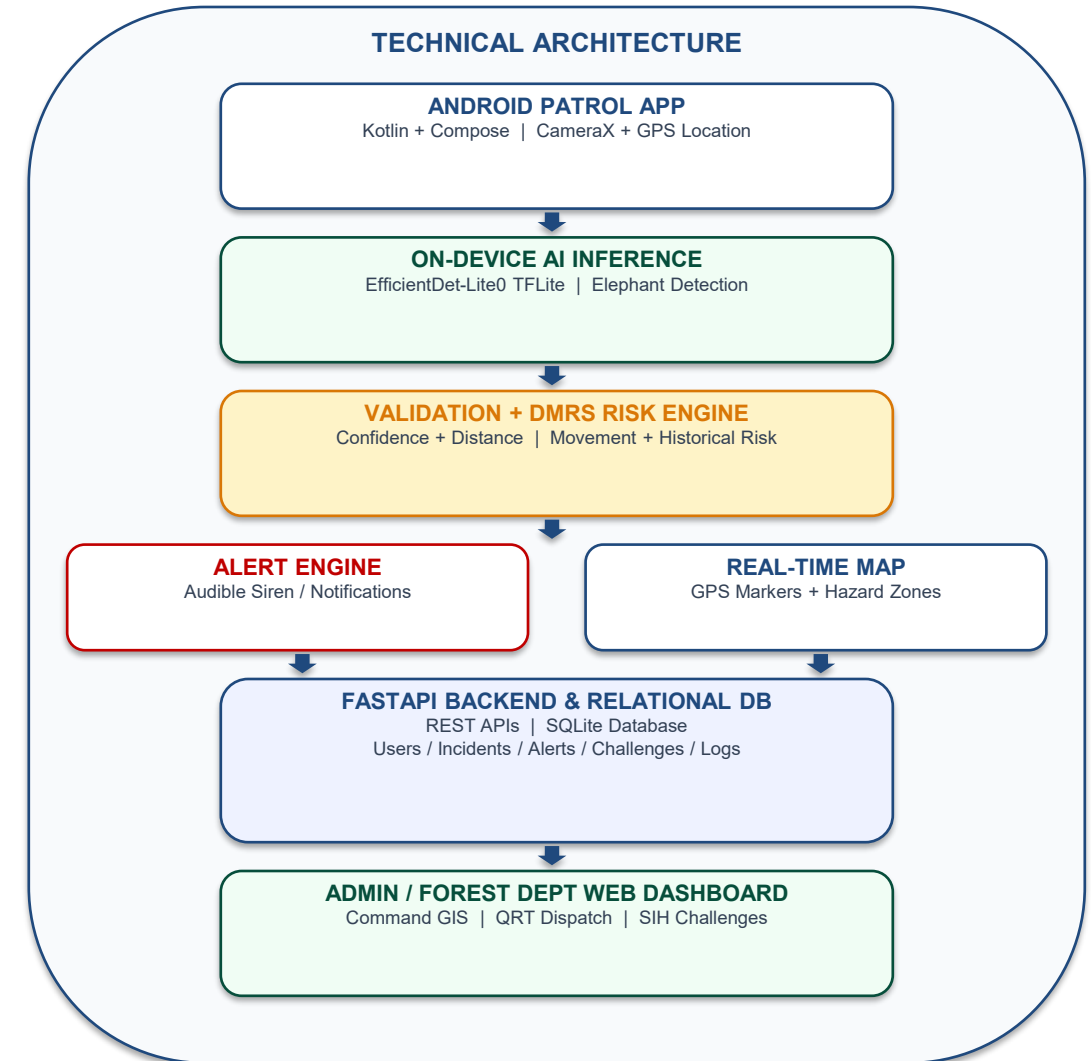
- FastAPI (Python 3.11) high-throughput asynchronous server
- RESTful API endpoints for telemetry and incident intake
- WebSocket & SSE event bus for real-time alert streaming

API & Synchronization:

- Telemetry reporting & incident/alert synchronization
- Bi-directional sync (Android ↔ Backend ↔ Admin Dashboard)
- Offline queue with automated background sync via WorkManager

Database & Governance:

- SQLite / Relational database for prototype backend
- Tables: Users, Incidents, Alerts, Challenges, Solutions, Audit Logs
- Parameterized queries and role-based access control (RBAC)



Feasibility-

Technical Feasibility:

- On-device lightweight AI reduces dependence on continuous cloud inference.
- Smartphone camera + GPS provide the primary sensing layer.
- FastAPI provides a lightweight central integration layer.
- Existing camera/telecom infrastructure can complement mobile nodes.

Operational Feasibility:

- Designed for forest-fringe and low-connectivity environments.
- Multi-channel warning architecture (app, siren, SMS) ensures wide reach.
- Forest Department dashboard supports centralized monitoring.
- Pilot deployment can begin in selected high-risk zones (Dalma NH-33).

Economic Feasibility:

- Built on open-source software frameworks with zero licensing fees.
- Uses commodity smartphones instead of specialized military hardware.
- Existing infrastructure can be reused where available.
- Modular architecture allows phased, cost-effective rollout.

Viability-

Social Viability:

- Early warnings reduce human-elephant encounters, protecting lives and crops.
- Connects local communities with Forest Department response teams.
- Builds trust through transparent, shared incident data and alerts.

Environmental Viability:

- Non-invasive camera monitoring does not disturb natural elephant movement.
- Helps eliminate retaliatory killings, poisonings, and electrocutions.
- Historical conflict data directly supports long-term conservation planning.
- Contributes to India's national wildlife conservation commitments (Project Elephant).

KEY CHALLENGES & ENGINEERED MITIGATION STRATEGIES

⚠ Detection Uncertainty

→ Temporal validation (3-of-5 queue) + confidence thresholding

⚠ Camera Coverage Gaps

→ Mobile sensing units + integration with existing infrastructure

⚠ Spam / Fake Reports

→ Incident verification + AI media checking + user trust scoring

⚠ Network Blackouts

→ On-device edge AI inference + offline-first sync architecture

➤ Benefits-

Social Benefits:

- Faster early warning for communities in elephant-prone corridors.
- Reduced risk of human casualties, injuries, and surprise encounters.
- Protects crops and rural property from devastating nocturnal raids.
- Accessible warning via multiple channels (app, audible siren, SMS).
- Strengthened coordination between citizens and Forest Department.

Technical Benefits:

- Demonstrates practical real-world edge AI vision for wildlife safety.
- Combines on-device AI, GPS proximity, risk scoring, and GIS mapping.
- Centralized incident and alert management with audit logging.
- Reusable framework adaptable to other conflicts (tigers, leopards, boars).

Environmental Benefits:

- Supports non-invasive wildlife monitoring without disturbing natural movement.
- Helps prevent retaliatory elephant killing, electrocution, and poisoning.
- Provides historical data for corridor identification and conservation planning.

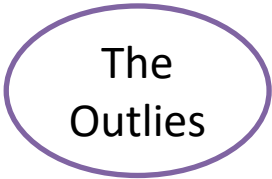
➤ Potential Impact-

1. IMPACT & INTERVENTION PIPELINE

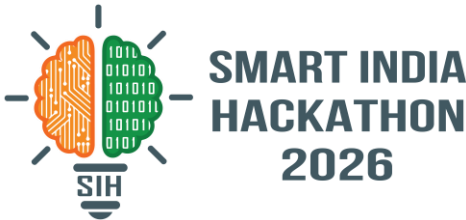


2. SIH COLLABORATIVE PROBLEM-SOLVING CYCLE





RESEARCH AND REFERENCES



Research & References:

- **Down To Earth:** State of India's Environment (human-wildlife conflict trends)
- **PMC / PubMed Central:** Human-elephant conflict studies (Jharkhand & Chhattisgarh)
- **Nature (Scientific Reports):** Fatality trends & mitigation strategies in corridor belts
- **LILA BC:** Open wildlife camera-trap image & benchmark datasets

Existing AI / Wildlife Platforms:

- **ElephantWatch (Amrita Univ):** AI-enhanced elephant detection system
- **EleSense (VFAES):** Sensor-based elephant corridor warning system
- **MegaDetector (Microsoft AI for Earth):** Open-source wildlife detection model

Competitive Analysis

Key Parameter	ElephantWatch	EleSense	Public Cameras	ELEPHANT GUARD
Real-time detection	✓ (Server)	✓ (Sensor)	— (Manual)	✓ (On-Device AI)
Risk-zone assessment	— (Basic)	— (Track)	— (None)	✓ (DMRS Engine)
Temporal validation	— (Single)	— (Threshold)	— (None)	✓ (3-of-5 Queue)
Proximity alerting	— (General)	✓ (Local)	— (None)	✓ (0-1km / 1-2.5km)
Forest Dept dashboard	✓ (Monitor)	— (Hardware)	— (Feed only)	✓ (Full Web GIS)
Crowdsourced reports	— (Sensor)	— (Sensor)	— (None)	✓ (Citizen + QRT)
University / Industry hub	— (Internal)	— (NGO)	— (None)	✓ (SIH Challenge)
Scalable to other wildlife	— (Elephant)	— (Elephant)	✓ (Manual)	✓ (Modular)